

**2015 HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper

Total marks – 100

Section I Pages 2–28

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–30
- Allow about 1 hour and 40 minutes for this part

Section II Pages 29–39

25 marks

- Attempt ONE question from Questions 31–35
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

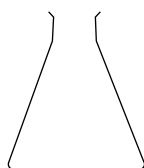
Use the multiple-choice answer sheet for Questions 1–20.

1 In which layer of the atmosphere does ozone absorb the most UV radiation?

- (A) Mesosphere
- (B) Stratosphere
- (C) Thermosphere
- (D) Troposphere

2 Which type of glassware is used in a titration to deliver an accurate volume of a solution to a known volume of another solution?

(A)



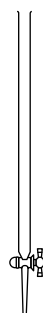
(B)



(C)



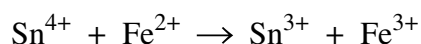
(D)



3 What flame colour do copper ions produce when heated?

- (A) Brick red
- (B) Blue-green
- (C) Pale purple
- (D) Yellow-orange

4 What happens to Fe^{2+} in the following reaction?



- (A) It undergoes oxidation and gains electrons.
- (B) It undergoes reduction and gains electrons.
- (C) It undergoes oxidation and loses electrons.
- (D) It undergoes reduction and loses electrons.

5 The oxides CaO , CO_2 , Na_2O and N_2O_4 are placed in water to form four separate solutions.

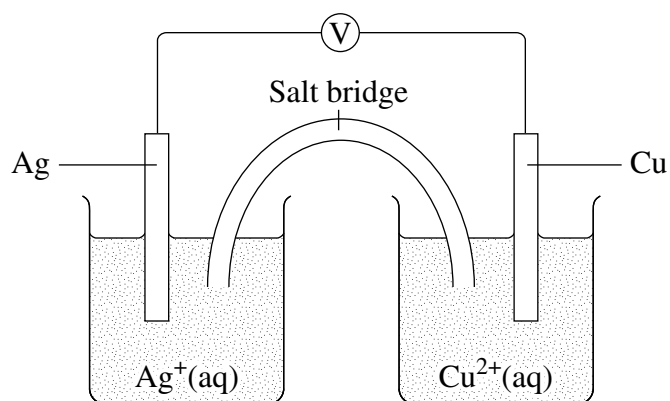
Which row of the table correctly indicates the solutions with pH less than 7 and the solutions with pH greater than 7?

<i>Solutions</i>				
<i>pH less than 7</i>			<i>pH greater than 7</i>	
(A)	CO_2	N_2O_4	CaO	Na_2O
(B)	CaO	N_2O_4	CO_2	Na_2O
(C)	CaO	Na_2O	CO_2	N_2O_4
(D)	CO_2	Na_2O	CaO	N_2O_4

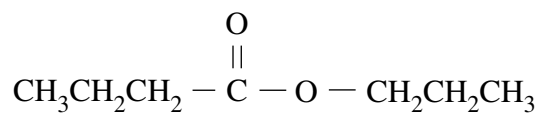
6 Which of the following is the most suitable replacement for CFCs in terms of reducing their environmental impact?

- (A) CH_4
- (B) CH_2F_2
- (C) CH_2ClF
- (D) $\text{CHCl}_2\text{CCl}_2\text{F}$

- 7 A diagram of a simple cell is shown.



- Which of the following occurs when the cell is in operation?
- (A) Silver ions are formed in solution.
(B) The copper electrode loses electrons.
(C) Electrons travel through the electrolyte.
(D) The copper electrode increases in mass.
- 8 Which of the following statements best explains the solubility of ethanol in octane?
- (A) Ethanol and octane are both non-polar.
(B) Ethanol forms hydrogen bonds with octane.
(C) Ethanol forms dispersion forces with octane.
(D) Ethanol forms dipole-dipole bonds with octane.
- 9 What are the reactants used to make this compound?

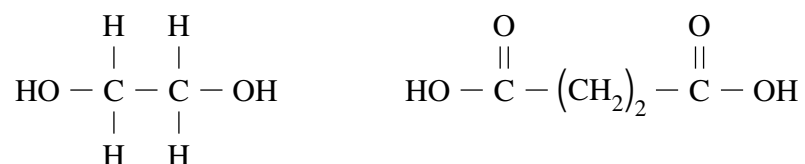


- (A) Butan-1-ol and butanoic acid
(B) Butan-1-ol and propanoic acid
(C) Propan-1-ol and butanoic acid
(D) Propan-1-ol and propanoic acid

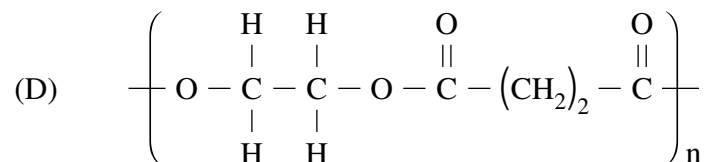
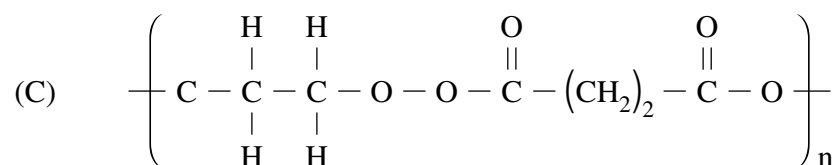
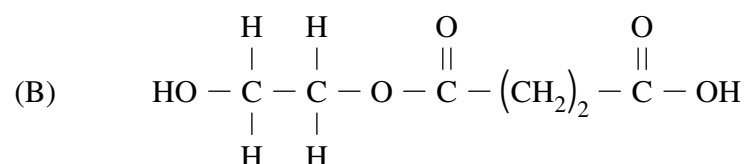
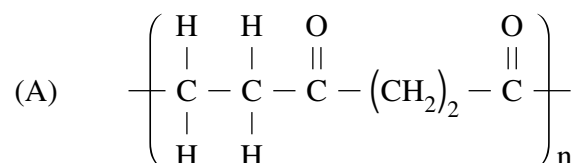
10 Which of the equations correctly describes incomplete combustion?

- (A) $\text{C}_2\text{H}_5\text{OH}(l) + 2\text{O}_2(g) \rightarrow 2\text{CO}(g) + 3\text{H}_2\text{O}(l)$
 (B) $\text{C}_2\text{H}_5\text{OH}(l) + \frac{7}{2}\text{O}_2(g) \rightarrow 2\text{CO}_2(g) + 3\text{H}_2\text{O}(l)$
 (C) $\text{C}_2\text{H}_5\text{OH}(l) + 3\text{O}_2(g) \rightarrow 2\text{CO}_2(g) + 3\text{H}_2\text{O}(l)$
 (D) $\text{C}_2\text{H}_5\text{OH}(l) + 2\text{O}_2(g) \rightarrow \text{C}(s) + \text{CO}(g) + 3\text{H}_2\text{O}(l)$

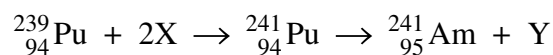
11 Two monomers are shown.



Which of the following shows a condensation polymer that could be formed from the monomers?



- 12 A transuranic element can be produced in a nuclear reactor according to this equation:



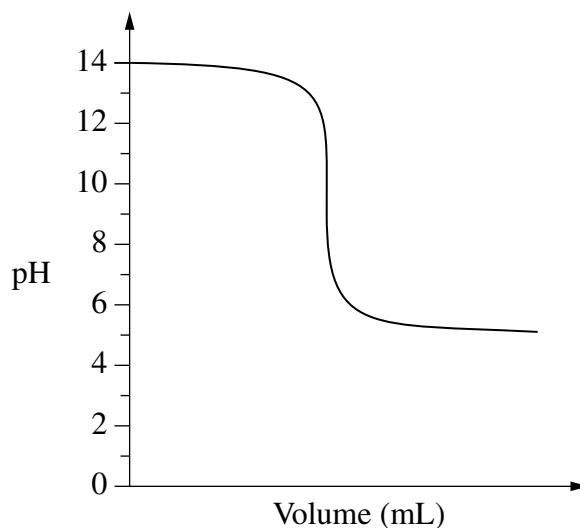
Which row of the table correctly identifies X and Y?

	X	Y
(A)	Neutron	Electron
(B)	Proton	Neutron
(C)	Neutron	Proton
(D)	Proton	Electron

- 13 Which of the following solutions has the highest pH?

- (A) 1.0 mol L⁻¹ acetic acid
- (B) 0.10 mol L⁻¹ acetic acid
- (C) 1.0 mol L⁻¹ hydrochloric acid
- (D) 0.10 mol L⁻¹ hydrochloric acid

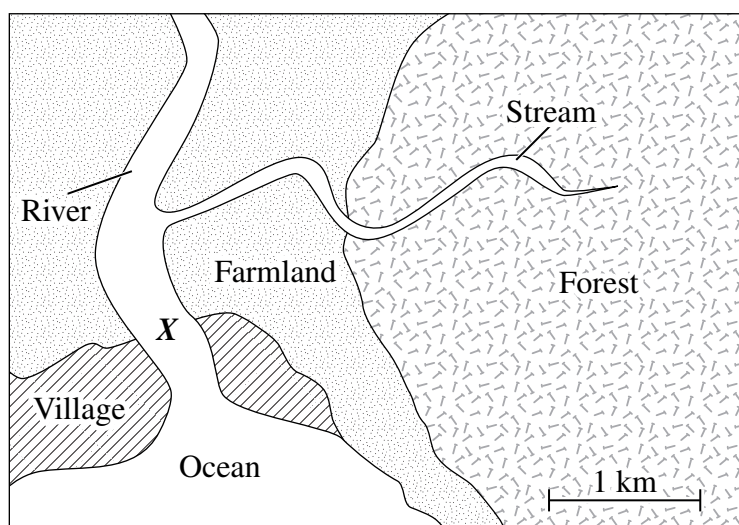
- 14 The graph shows the changes in pH during a titration.



Which pH range should an indicator have to be used in this titration?

- (A) 3.1–4.4
- (B) 5.0–8.0
- (C) 6.0–7.6
- (D) 8.3–10.0

- 15 Part of a water catchment is shown in the diagram.

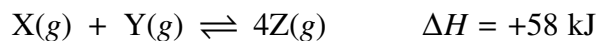


A sample of river water taken from point *X* is analysed.

Which row of the table shows the most likely results?

Results of water analysis at <i>X</i>				
	<i>Turbidity</i> (NTU)	<i>BOD</i> (ppm)	<i>pH</i>	<i>Total dissolved solids</i> (ppm)
(A)	400	18	6.5	22 000
(B)	22	3	8.5	17
(C)	5	18	6.5	22 000
(D)	400	3	8.5	17

- 16 The equation describes an equilibrium reaction occurring in a closed system.



Under which set of conditions would the highest yield of *Z(g)* be obtained?

	<i>Temperature</i> (°C)	<i>Pressure</i> (kPa)
(A)	50	100
(B)	50	200
(C)	300	100
(D)	300	200

- 17** What volume of carbon dioxide will be produced if 10.3 g of glucose is fermented at 25°C and 100 kPa?
- (A) 1.30 L
(B) 1.42 L
(C) 2.57 L
(D) 2.83 L

Use this information to answer Questions 18–19.

A sample of pond water from a contaminated site was analysed to determine the concentration of lead ions using the following procedure.

- A measuring cylinder was used to collect a 50 mL sample from the pond.
- The sample was placed in a clean dry beaker.
- 25.0 mL of 0.200 mol L⁻¹ sodium chloride solution was added to the sample.
- The precipitate of lead(II) chloride that formed was filtered, dried and weighed. It had a mass of 0.13 g.

- 18** How could the reliability of the analysis of the pond water be improved?
- (A) Analyse more samples from the same pond
(B) Use 50 mL of distilled water as a control sample
(C) Analyse samples from different ponds on the site
(D) Remove other contaminants from the sample before the analysis
- 19** What was the concentration of lead ions in the sample?
- (A) 5.0×10^{-3} mol L⁻¹
(B) 5.8×10^{-3} mol L⁻¹
(C) 9.3×10^{-3} mol L⁻¹
(D) 10.7×10^{-3} mol L⁻¹

- 20 The table shows the heat of combustion of four straight chain alkanols.

<i>Number of C atoms in straight chain alkanol</i>	<i>Heat of combustion (kJ mol⁻¹)</i>
1	726
3	2021
5	3331
7	4638

What is the mass of water that could be heated from 20°C to 45°C by the complete combustion of 1.0 g of heptan-1-ol?

- (A) 0.032 kg
- (B) 0.044 kg
- (C) 0.36 kg
- (D) 0.38 kg

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Centre Number

Section I (continued)

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Student Number

Part B – 55 marks**Attempt Questions 21–30****Allow about 1 hour and 40 minutes
for this part**

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided on pages 27 and 28. If you use this space, clearly indicate which question you are answering.

Write your Centre Number and Student Number at the top of this page.

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Please turn over

Question 21 (4 marks)

- (a) Outline a suitable method to prepare a natural indicator.

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- (b) How could a natural indicator be tested?

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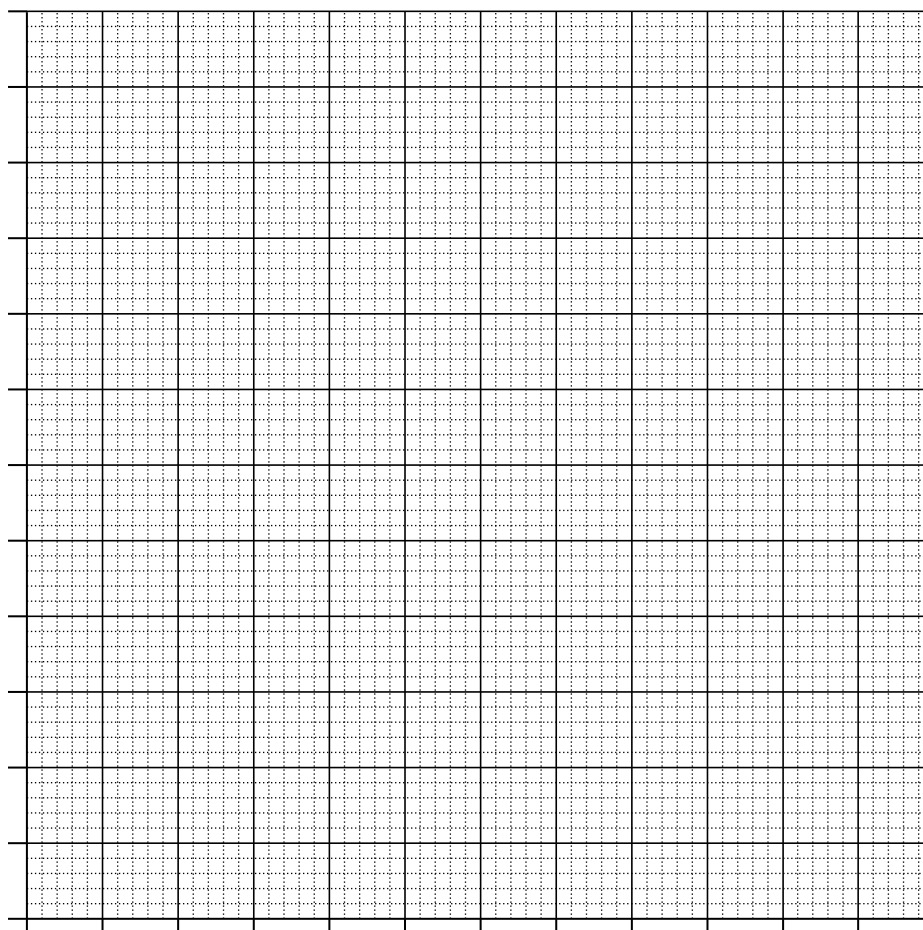
Question 22 (7 marks)

The table shows data for ozone concentrations over 50 years in the upper atmosphere above Antarctica.

<i>Year</i>	<i>Ozone Concentration</i> (Dobson Units)
1955	320
1960	300
1970	300
1980	260
1995	130
2000	130
2005	150

- (a) Draw a line graph of the data on the grid provided.

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Question 22 continues on page 16

Question 22 (continued)

(b) Describe a method by which this data could have been measured.

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End of Question 22

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Question 23 (4 marks)

Explain how the structure and chemistry of ONE of the following cells determines its cost and practicality.

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- button cell
- fuel cell
- vanadium redox cell
- lithium cell
- liquid junction photovoltaic device (eg the Gratzel cell)

Name of cell:

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Question 24 (5 marks)

- (a) Explain why the salt, sodium acetate, forms a basic solution when dissolved in water. Include an equation in your answer. **2**

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- (b) A solution is prepared by using equal volumes and concentrations of acetic acid and sodium acetate. **3**

Explain how the pH of this solution would be affected by the addition of a small amount of sodium hydroxide solution. Include an equation in your answer.

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Question 25 (7 marks)

- (a) Describe the steps involved in the process of *addition polymerisation*.

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- (b) Explain the uses of polyethylene and polystyrene in terms of their structures and properties.

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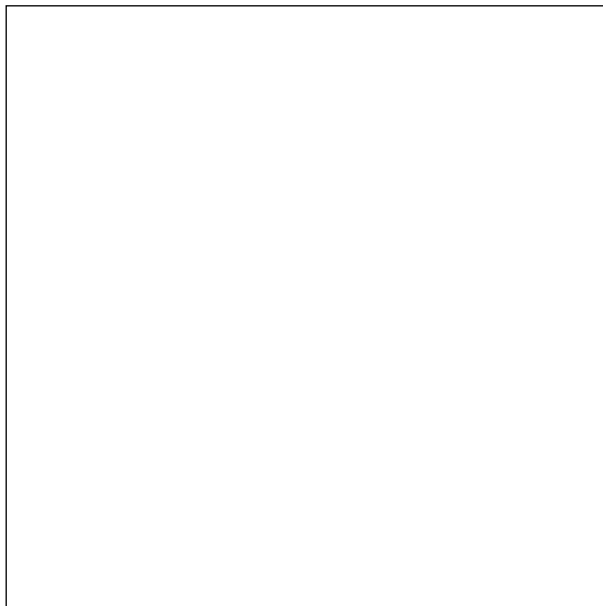
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Question 26 (7 marks)

A sodium hydroxide solution was titrated against citric acid ($\text{C}_6\text{H}_8\text{O}_7$) which is triprotic.

- (a) Draw the structural formula of citric acid
(2-hydroxypropane-1,2,3-tricarboxylic acid).

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- (b) How could a computer-based technology be used to identify the equivalence point of this titration?

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Question 26 continues on page 21

Question 26 (continued)

- (c) The sodium hydroxide solution was titrated against 25.0 mL samples of 0.100 mol L⁻¹ citric acid. The average volume of sodium hydroxide used was 41.50 mL.

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Calculate the concentration of the sodium hydroxide solution.

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End of Question 26

Question 27 (5 marks)

Name a radioisotope used in a non-medical industry and discuss its use in that industry in terms of its properties.

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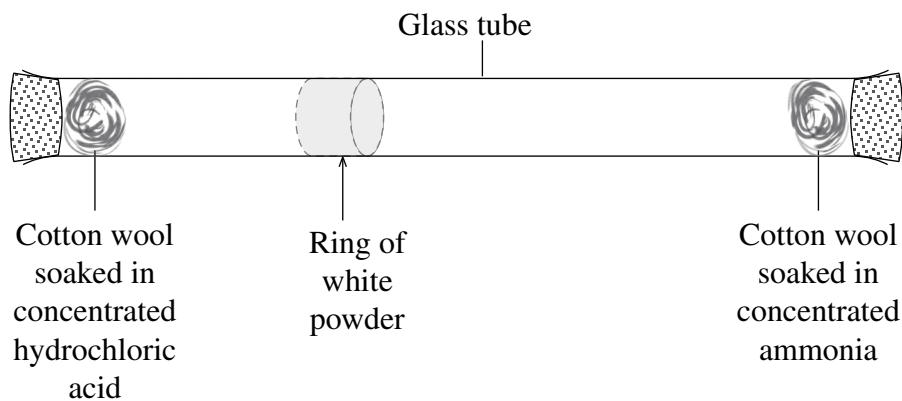
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Question 28 (3 marks)

The equipment shown is set up. After some time a ring of white powder is seen to form on the inside of the glass tube.



- (a) Why would this NOT be an acid–base reaction according to Arrhenius? 1

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- (b) Explain why this would be considered a Brønsted–Lowry acid–base reaction. Include an equation in your answer. 2

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Question 29 (7 marks)

The procedure of a first-hand investigation conducted in a school laboratory to determine the percentage of sulfate in a lawn fertiliser is shown.

- 2.00 g of a sample of fertiliser was ground up and placed in a beaker.
- It was dissolved in about 200 mL of 0.1 mol L⁻¹ hydrochloric acid, stirred and filtered.
- Excess barium chloride solution was quickly added to this beaker and a precipitate formed.
- The precipitate was then allowed to settle, filtered using filter paper and the residue collected.
- The residue was dried and weighed and had a mass of 2.23 g.

- (a) Suggest modifications that could be made to the procedure to improve the results of this investigation. Justify your suggestions.

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Question 29 continues on page 25

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Question 29 (continued)

- (b) Calculate the percentage of sulfate in the original fertiliser sample. 3

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End of Question 29

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Section I Part B extra writing space

If you use this space, clearly indicate which question you are answering.

Handwriting practice area with horizontal lines.

Section I Part B extra writing space

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 31–35

Allow about 45 minutes for this section

Answer parts (a)–(d) of one question in the Section II Writing Booklet. Extra writing booklets are available.

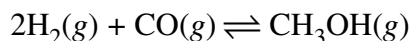
Show all relevant working in questions involving calculations.

	Pages
Question 31 Industrial Chemistry	30–31
Question 32 Shipwrecks, Corrosion and Conservation	32–33
Question 33 The Biochemistry of Movement	34–35
Question 34 The Chemistry of Art	36–37
Question 35 Forensic Chemistry	38–39

Question 31 — Industrial Chemistry (25 marks)

Answer parts (a) and (b) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) At temperatures above 100°C, hydrogen and carbon monoxide react to form methanol gas in this reversible reaction.



A mixture of hydrogen, carbon monoxide and methanol is placed in a container with a volume that can be changed. The mixture is allowed to reach equilibrium.

- (i) The initial volume of the container is 1.00 L. **2**

Account for any changes in the concentration of hydrogen gas when the volume of the container is rapidly increased to 2.00 L.

- (ii) The initial mixture placed in the container had 0.50 mol of hydrogen, 1.00 mol of carbon monoxide and 2.50 mol of methanol. Once the volume of the container had been increased to 2.00 L and equilibrium had been re-established, the number of moles of hydrogen in the mixture had changed by 0.36 mol. **3**

Calculate the equilibrium constant for this reaction.

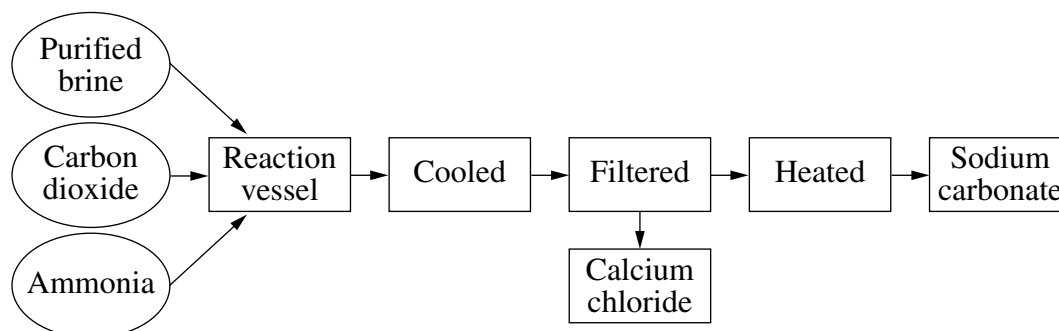
- (b) (i) Describe how saponification can be safely carried out as part of a first-hand investigation. **3**
- (ii) Explain the chemistry related to the cleaning properties of the product of saponification. **4**

Question 31 continues on page 31

Question 31 (continued)

Answer parts (c) and (d) of the question on pages 5–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (c) The diagram shows part of the Solvay process for producing sodium carbonate.



- (i) Outline the chemistry of the production of sodium carbonate in the process shown. Include equations in your answer. **3**
- (ii) By making specific reference to the diagram, justify the requirements for the location of a Solvay process plant. **3**
- (d) Compare the membrane cell method with ONE other method used in the industrial production of sodium hydroxide in terms of technical and environmental issues. **7**

End of Question 31

Question 32 — Shipwrecks, Corrosion and Conservation (25 marks)

Answer parts (a) and (b) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

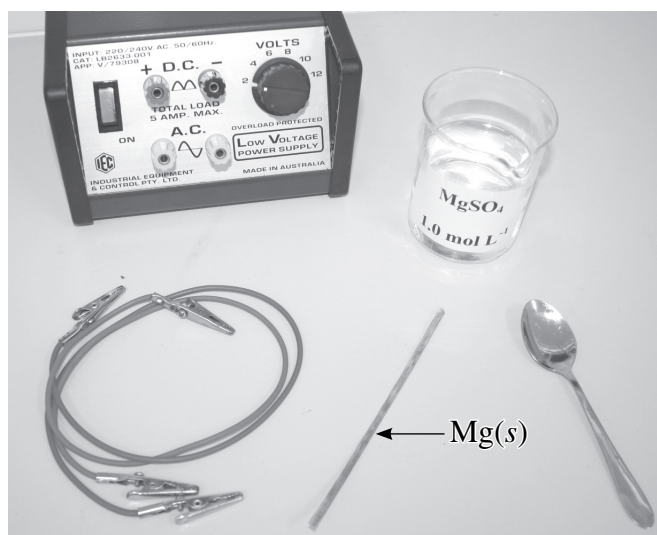
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|-----|------|---|----------|
| (a) | (i) | Outline the limitations of using paint to protect ships that are in constant use. | 2 |
| | (ii) | Explain the chemical principles involved in the use of a sacrificial anode. Include relevant chemical equations in your answer. | 3 |
| (b) | (i) | Describe a valid and reliable first-hand investigation that can be used to compare the rates of corrosion of iron with ONE named form of steel. | 3 |
| | (ii) | Explain how the percentage composition of steel can determine its properties, with reference to TWO types of steel. | 4 |

Question 32 continues on page 33

Question 32 (continued)

Answer parts (c) and (d) of the question on pages 5–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (c) (i) The equipment in the photograph was used in an attempt to plate a metal spoon with magnesium using an electrolytic cell containing a solution of magnesium sulfate. 3



Draw a labelled scientific diagram of the electrolytic cell. Include the cathode, anode, direction of electron flow and polarity of the electrodes.

- (ii) Explain how Davy's work increased our understanding of electron transfer reactions. 3
- (d) Two identical ships are sunk in seawater. One is sunk in shallow water (60 m) and the other in deep water (4000 m). Explain how the rusting processes differ in these two ships. Include equations in your answer. 7

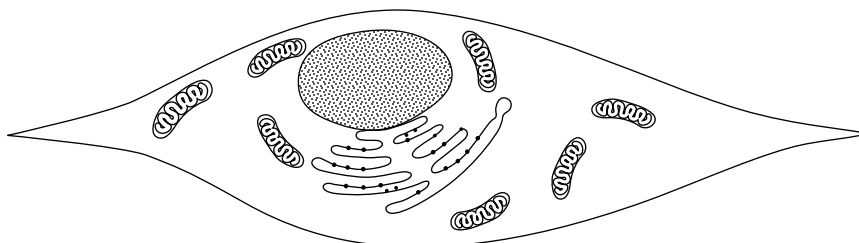
End of Question 32

Question 33 — The Biochemistry of Movement (25 marks)

Answer parts (a) and (b) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) A simplified diagram of a muscle cell is shown.

2



Identify the TWO components of the cell that are involved in respiration AND the type of respiration that occurs in each.

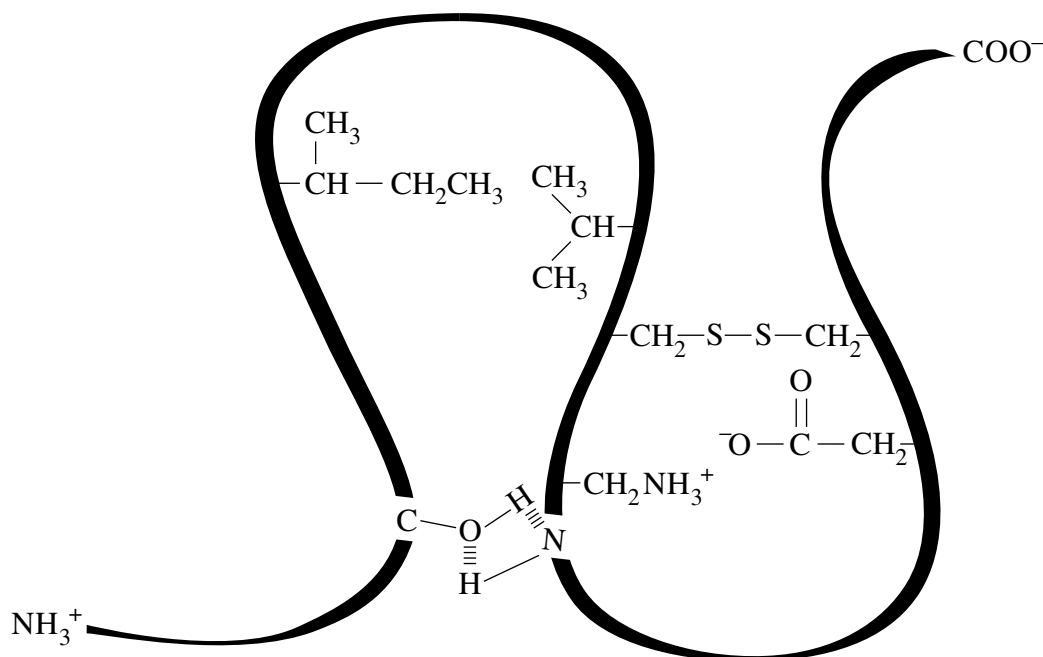
- (ii) Explain the biological significance of adenosine triphosphate (ATP) with reference to its structure. **3**
- (b) (i) Describe a reliable and valid procedure that can be carried out to demonstrate the effect of changing temperature on enzyme function. **3**
- (ii) Discuss the use of models in understanding how enzymes function in living systems. **4**

Question 33 continues on page 35

Question 33 (continued)

Answer parts (c) and (d) of the question on pages 5–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (c) A section of a protein is shown.



- (i) Using TWO examples from the diagram, explain how bonding between sections of the protein chain influences its shape. 3
- (ii) With reference to the diagram, account for ways in which this protein may be denatured. 3
- (d) Compare the metabolic pathways involved in sprinting with those involved when walking at a gentle pace. 7

End of Question 33

Question 34 — The Chemistry of Art (25 marks)

Answer parts (a) and (b) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

- (a) (i) Identify the chemical composition of a cosmetic used in an ancient culture and identify the potential health risk associated with the use of the cosmetic. **2**
- (ii) Explain why $\text{Cr}^{2+}(\text{aq})$ is coloured whereas $\text{Zn}^{2+}(\text{aq})$ is not coloured. **3**
- (b) An investigation is to be conducted to observe the flame colour of some metal ions.
- (i) Describe a safe and valid procedure that can be used to carry out this investigation, identifying the specific metal ions observed. **3**
- (ii) Explain why only certain metal ions can be identified using flame colours, naming ONE metal ion that cannot be identified using flame colours. **4**

Question 34 continues on page 37

Question 34 (continued)

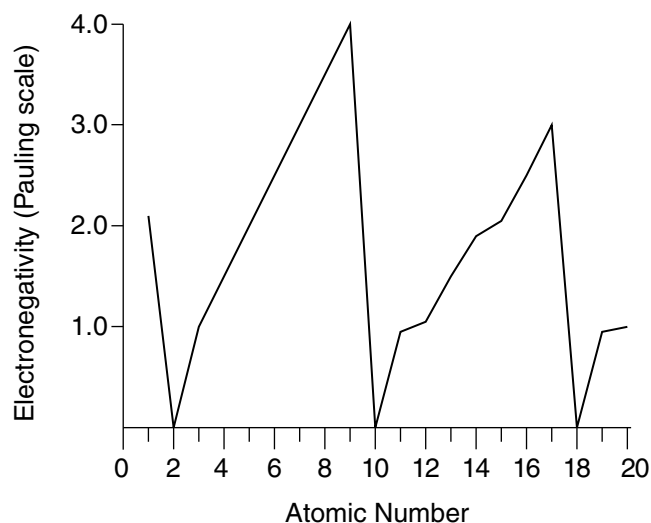
Answer parts (c) and (d) of the question on pages 5–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (c) (i) The first ionisation energy of each of the elements in the third row of the Periodic Table is given. **3**

Element	Na	Mg	Al	Si	P	S	Cl	Ar
First ionisation energy (kJ mol ⁻¹)	494	736	577	786	1060	1000	1260	1520

Use the data in the table to explain how the first ionisation energy of these elements can provide evidence for the existence of sub-shells in atoms.

- (ii) The graph shows the electronegativity values for elements in Periods 1, 2 and 3 of the Periodic Table.



Use the graph to explain the relationship between the arrangement of electrons in an element and its electronegativity. **3**

- (d) Assess how the Bohr model of the atom has contributed to our understanding of atomic structure, making reference to emission spectra. **7**

End of Question 34

Question 35 — Forensic Chemistry (25 marks)

Answer parts (a) and (b) of the question on pages 2–4 of the Section II Writing Booklet. Start each part of the question on a new page.

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|-----|------|--|----------|
| (a) | (i) | Compare the composition of glycogen with that of cellulose. | 2 |
| | (ii) | Relate the differences in composition of glycogen and cellulose to their different structures. | 3 |
| (b) | (i) | Identify the structure of amino acids and describe the relationship between amino acids and proteins. | 3 |
| | (ii) | Describe a safe and valid procedure that can be used to show the presence of protein in egg white. Include expected results. | 4 |

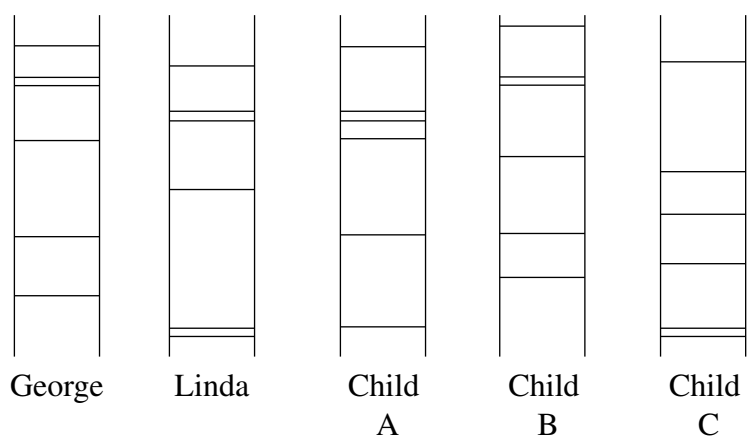
Question 35 continues on page 39

Question 35 (continued)

Answer parts (c) and (d) of the question on pages 5–8 of the Section II Writing Booklet. Start each part of the question on a new page.

- (c) (i) George and Linda have one child together. Each of them also has one child from a previous relationship. A schematic representation of their DNA profiles is shown below. 3

DNA profiles obtained from George, Linda and the three children



Use the information in the DNA profiles to identify the relationships of Child A, Child B and Child C to George and Linda. Justify your answer.

- (ii) Describe the benefits of maintaining DNA data banks. 3
- (d) Name ONE chromatography technique and assess its use in the analysis of forensic evidence. 7

End of paper

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Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -mC\Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium	
3 Li 6.941 Lithium	4 Be 9.012 Beryllium	Atomic Number Symbol		79 Au 197.0 Gold		Atomic Number Symbol		10 Ne 20.18 Neon		9 F 19.00 Fluorine		8 O 16.00 Oxygen	
		Standard Atomic Weight Name		197.0 Gold		197.0 Gold		16.00 Oxygen		19.00 Fluorine		16.00 Oxygen	
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	Lanthanoids		73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead
87 Fr Francium	88 Ra Radium	Actinoids	Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium		
		Actinoids											

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Elements with atomic numbers 113 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.

2015 HSC Chemistry Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	B
2	D
3	B
4	C
5	A
6	B
7	B
8	C
9	C
10	A
11	D
12	A
13	B
14	D
15	A
16	C
17	D
18	A
19	C
20	D

Section I, Part B**Question 21 (a)**

Criteria	Marks
• Correctly outlines all steps	2
• Identifies some steps	1

Sample answer:

- Collect some coloured plant material such as red cabbage. Select the coloured parts and cut into small pieces
- Place them in a beaker with water and boil until the water becomes coloured
- Cool mixture
- Decant the liquid into a container, leaving the solid behind.

Question 21 (b)

Criteria	Marks
• Outlines a suitable activity using a natural indicator to show the different colours in acidic and basic solutions	2
• Identifies a suitable activity for testing a natural indicator	1

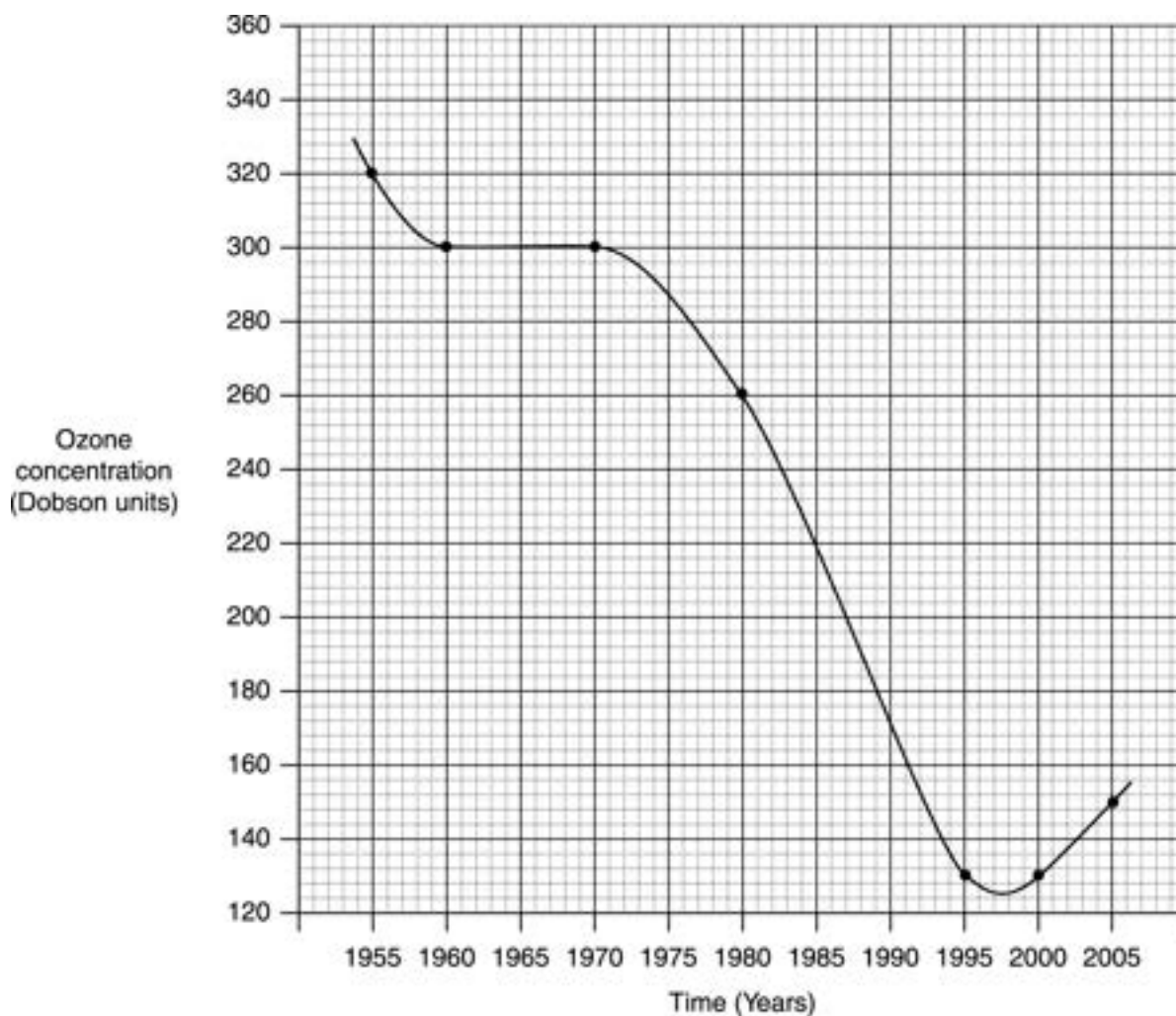
Sample answer:

Add some of the indicator to separate test tubes of a known acid and base.

Record the natural indicator colour in the acid and base.

Question 22 (a)

Criteria	Marks
• Provides a correct graph that includes labelled axes, units, appropriate scales, plotted points and a smooth line of best fit	4
• Provides a substantially correct graph	3
• Provides a graph that includes a scale	2
• Provides some relevant information	1

Sample answer:

Question 22 (b)

Criteria	Marks
• Describes how a ground-based technique or instrument launched into the upper atmosphere can be used to measure ozone levels	3
• Outlines an instrument and/or a technique that can be used to measure ozone levels	2
• Identifies an instrument or a technique that can be used to measure ozone levels	1

Sample answer:

The information could be measured by a ground-based UV spectrophotometer. This analyses UV light intensity at a wavelength at which ozone absorbs, and compares the intensity to a nearby wavelength of UV that ozone does not absorb. The amount of absorption by ozone provides information about ozone concentrations. Similar measurements over time allow ozone concentrations to be monitored.

Question 23

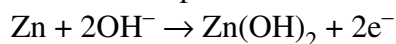
Criteria	Marks
• Outlines the structure and chemistry of the cell • Relates these to its cost and practicality	4
• Outlines the structure and chemistry of the cell • Links these to its cost or practicality OR • Outlines some features of the structure and/or the chemistry of the cell • Links these to its cost and practicality	3
• Outlines some features of the cell's structure and/or its chemistry and/or its cost and/or its practicality	2
• Provides some relevant information about the cell, its cost or its practicality	1

Sample answer:

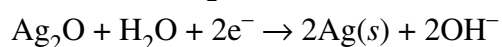
Name of cell: *Silver Button Cell*

Chemistry:

Anode: Zinc powdered



Cathode: C/Ag₂O paste



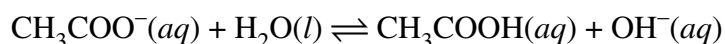
Electrolyte: KOH/Zn(OH)₂ paste

Since the cell is small in size, lightweight and has a stable voltage over a long period of time it is useful in small appliances such as cameras, calculators and watches.

Whilst silver is an expensive metal, the cell is relatively in expensive to purchase, these days.

Question 24 (a)

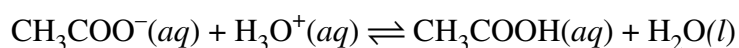
Criteria	Marks
- Recognises that the salt produces hydroxide ions and relates this to the formation of a basic solution - Includes a relevant equation	2
- Recognises that hydroxide ions are produced OR - Includes a relevant equation	1

Sample answer:

The presence of OH^- ions produced by the hydrolysis of CH_3COO^- increases the pH of the solution and results in a basic pH.

Question 24 (b)

Criteria	Marks
- Explains how an increase in $[\text{OH}^-]$ will affect the reaction - Relates to minimal change in pH - Includes a relevant equation	3
- Provides some explanation and/or description of how the pH would be affected - Links the reaction of OH^- to the equilibrium reaction	2
Provides some relevant information	1

Sample answer:

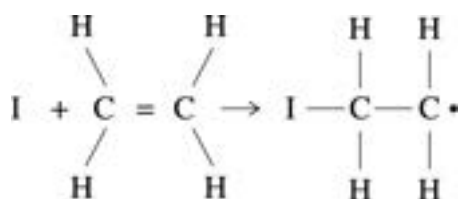
The addition of OH^- ions will cause reaction with H_3O^+ ions, reducing their concentration in the equilibrium mixture. This will force the reaction to the left to increase the $[\text{H}_3\text{O}^+]$, thus minimising the change in pH.

Question 25 (a)

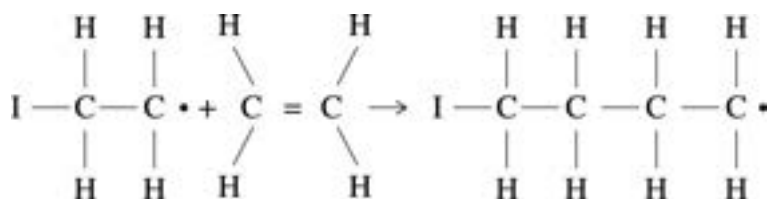
Criteria	Marks
• Includes the main features of the steps of addition polymerisation	3
• Identifies the majority of the main features of addition polymerisation	2
• Identifies ethylene as a monomer OR • Identifies a relevant step OR • Identifies a relevant chemical concept	1

Sample answer:

Initiator (I) (organic peroxide) attacks ethylene molecule, breaking double bond, and producing an unpaired reactive electron on the end of the growing chain (initiation).



This will attack another ethylene molecule, increasing the length of the growing chain (propagation).



Chain length continues to grow in this fashion until two growing chains combine (termination).

Question 25 (b)

Criteria	Marks
• Relates the uses of both polymers to their structures and properties	4
• Relates the uses of both polymers to their structures and/or properties	3
• Gives the structures of both polymers, and a property/use of one of the polymers OR • Gives the properties of both polymers, and a structure/use of one of the polymers OR • Gives the uses of both polymers, and a structure/property of one of the polymers	2
• Gives the structure OR a use OR a property of one of the polymers	1

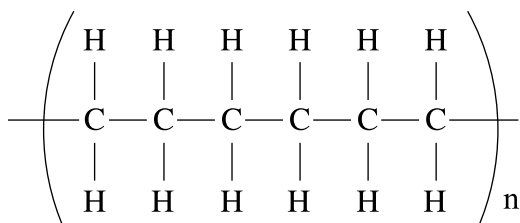
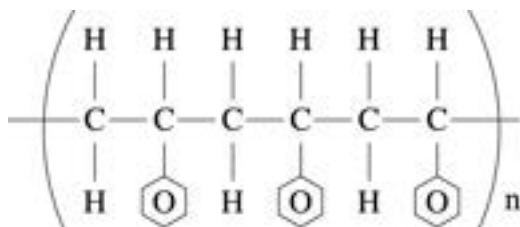
Sample answer:

Polystyrene consists of long carbon chains with benzene rings attached along the chain.

Polystyrene can be used for screwdriver handles and car battery cases because the large benzene rings create a much stiffer, more rigid structure.

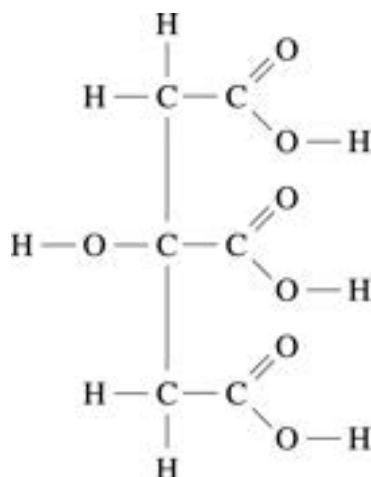
Polyethylene consists of long carbon chains with no side groups.

Polyethylene is used in cling-wrap and milk bottles because the long chains are flexible.

Answers could include:

Question 26 (a)

Criteria	Marks
• Draws a correct structural formula for citric acid	1

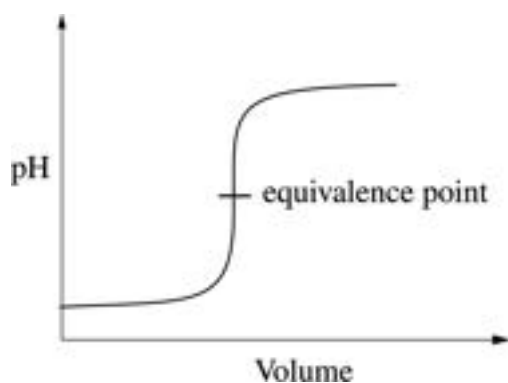
Sample answer:**Question 26 (b)**

Criteria	Marks
• Outlines how a computer-based technology could be used to identify the equivalence point	2
• Provides some relevant information	1

Sample answer:

A digital pH probe was used to collect appropriate data to plot a graph of pH vs volume of sodium hydroxide.

The equivalence point was identified from the graph obtained from the data collected by the pH probe.

Answer could include:

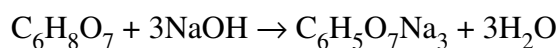
Question 26 (c)

Criteria	Marks
- Correctly calculates the concentration of sodium hydroxide - States answer to three significant figures	4
Correctly calculates the concentration of sodium hydroxide	3
- Applies correct steps to calculate the concentration of the sodium hydroxide solution OR - Correctly calculates moles of citric acid and moles of sodium hydroxide with correct equation	2
- Substitutes into a relevant formula OR - Provides a relevant equation OR - Shows a basic understanding of the calculation required	1

Sample answer:

$$n = cV = 0.100 \times 0.0250$$

$$\text{Moles citric acid} = 0.00250$$



$$41.50 \text{ mL NaOH must contain } 3 \times 0.00250 \text{ moles} \\ = 0.00750 \text{ moles}$$

$$c = \frac{n}{V} = \frac{0.00750}{0.04150} = 0.18072$$

$$= 0.181 \text{ mol L}^{-1} \text{ (3 significant figures)}$$

Question 27

Criteria	Marks
- Names a radioisotope used in a non-medical industry - Identifies issues and provides points for and/or against its use in that industry in terms of its properties	5
- Names a radioisotope used in a non-medical industry - Outlines its use in that industry and its properties relevant to the use	4
- Names a radioisotope used in a non-medical industry - Outlines its use in that industry and at least one of its properties relevant to the use	3
- Names a radioisotope used in a non-medical industry and identifies a use or a property of the radioisotope OR - Identifies use(s) and/or property(ies) of radioisotopes used in non-medical industries	2
- Names a radioisotope OR - Identifies a use or a property of radioisotopes used in non-medical industries	1

Sample answer:

Co-60 is used in industrial radiography to inspect metals and welds for defects

Beams of radiation are passed through the object to be checked from a selected source of Co-60. More radiation will pass through where there are cracks or weaknesses.

Co-60 is used as it is a gamma-emitter and penetrates metal, providing a non-destructive, fast, accurate and relatively cheap process. It also has a relatively long half-life (5.3 years), and therefore the radioisotope does not need to be replaced on a regular basis.

One disadvantage of working with Co-60 is that it is radioactive and beams of radiation are used to detect defects. It can be dangerous when workers are exposed to it, therefore care needs to be taken to minimise the risk.

Question 28 (a)

Criteria	Marks
Correctly shows why this would not be considered an Arrhenius acid-base reaction	1

Sample answer:

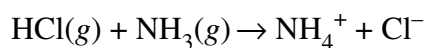
This reaction does not occur in aqueous solution.

Question 28 (b)

Criteria	Marks
- Provides a valid reason - Includes a relevant equation	2
Provides some relevant information	1

Sample answer:

It involves proton transfer.



Question 29 (a)

Criteria	Marks
• Provides appropriate suggestions with well-linked justifications	4
• Provides appropriate suggestions and justification for at least one of the suggestions	3
• Provides appropriate suggestions OR • Provides an appropriate suggestion with justification	2
• Provides an appropriate suggestion	1

Sample answer:

Barium sulfate is a very fine precipitate. After the BaCl_2 was added, the mixture could be heated for some time. This allows the fine BaSO_4 precipitate to flocculate into larger particles, which would be trapped by the filter. The pores of most filter paper are still too large to effectively capture the BaSO_4 precipitate, and so filtering should be done using a sintered glass crucible, which has small pore sizes.

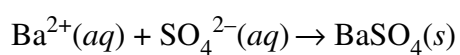
The sulfate will not be evenly distributed throughout the mixture, so to achieve more reliable and valid results, several samples of the fertiliser should be used so that an average value for the sulfate content in the mixture could be determined.

Question 29 (b)

Criteria	Marks
• Correctly calculates the percentage of sulfate in the original sample	3
• Provides a substantially correct response	2
• Applies a relevant step OR • Provides a relevant equation	1

Sample answer:

Calculations:



mass of fertiliser used: 2.00 g

mass of precipitate formed: 2.23 g

$$\text{mass of SO}_4^{2-} = \frac{\text{Molar mass of SO}_4^{2-}}{\text{Molar mass of BaSO}_4} \times 2.23 = \frac{96.07}{233.37} \times 2.23 = 0.918$$

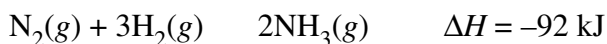
$$\therefore \% \text{ of SO}_4^{2-} \text{ in the fertiliser} = \frac{0.918}{2.00} \times 100 = 45.9\%$$

Question 30

Criteria	Marks
- Relates the conditions used in the Haber process to yield and rate considerations and Le Chatelier's principle - Relates information in the graph to these conditions - Provides a logical response with no extraneous information	6
- Links conditions used in the Haber process to yield, rate and Le Chatelier's principle - Links the graph to temperature and pressure	5
- Links conditions used in the Haber process to yield/rate and Le Chatelier's principle - Refers to the graph	4
Links conditions used in the Haber process to Le Chatelier's principle and/or yield and/or rate and/or the graph	3
Identifies condition(s) used in the Haber process and/or yield and/or rate and/or the graph and/or feature(s) of Le Chatelier's principle	2
Provides some relevant information	1

Sample answer:

In the Haber process, ammonia is produced from nitrogen and hydrogen in the following exothermic reaction.



Being exothermic, yield is increased by lowering the temperature of the reaction mixture, which would drive the reaction to the right, as according to Le Chatelier's principle, the reaction would be favoured that produced heat. On the graph, this is clearly shown, as for each pressure, the yield increased at lower temperatures. Maximum yield could be obtained at very low temperatures, however, the rate of ammonia formation would be compromised at low temperatures, hence, in the Haber process, a temperature of about 450°C is used. This produces an acceptable yield. A catalyst of Fe_3O_4 is used to maintain an acceptable rate of reaction at this low industrial temperature. In the reaction above, there is a 4:2 ratio of gaseous reactants to products, so higher pressures will drive the reaction to the right to reduce moles of gas, which reduces the pressure. Ideally as shown on the graph, extreme pressures would increase the yield. However, in the Haber process, the cost of maintaining high-pressure reaction vessels is prohibitive, so a pressure of 300 atm is used which is an acceptable compromise.

Section II

Question 31 (a) (i)

Criteria	Marks
• Accounts for changes in concentration	2
• Outlines a change in concentration	1

Sample answer:

Initially, $[H_2]$ would drop as the volume of the container was suddenly increased. However, as there is a 3:1 ratio of gas particles in the equilibrium reaction, the reverse reaction would be favoured to increase pressure by increasing the number of particles, and so $[H_2]$ would then start to increase again until equilibrium was re-established.

Question 31 (a) (ii)

Criteria	Marks
• Correctly calculates the equilibrium constant	3
• Provides some relevant steps	2
• Applies data to a relevant calculation	1

Sample answer:

	Moles H_2	Moles CO	Moles CH_3OH
<i>Initially</i>	0.50	1.00	2.50
<i>Used/made</i>	0.36	0.18	0.18
<i>Equilibrium</i>	0.86	1.18	2.32

$\therefore$ equilibrium concentrations in the 2.00 L container

$$[H_2] = 0.43$$

$$[CO] = 0.59$$

$$[CH_3OH] = 1.16$$

$$K = \frac{[CH_3OH]}{[H_2]^2 [CO]} = \frac{1.16}{0.43^2 \times 0.59} = 11$$

Question 31 (b) (i)

Criteria	Marks
- Provides a logical sequence of steps with relevant chemicals - Includes appropriate apparatus - Addresses a safety issue	3
- Provides some steps - Provides relevant chemicals and/or addresses a safety issue	2
- Includes an appropriate chemical OR - Provides a relevant step OR - Identifies a hazard OR - Includes a safety measure	1

Sample answer:

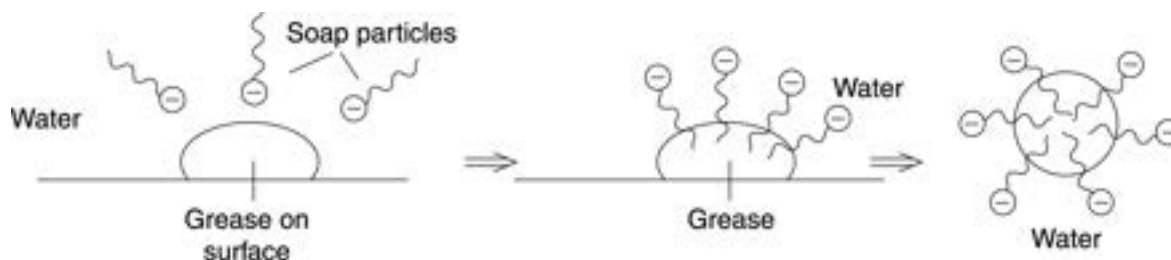
- Safety goggles used due to corrosive NaOH used in preparation
- 5 mL olive oil mixed with 5 mL ethanol and 5 mL NaOH solution in an evaporating basin
- Mixture heated over a hot water bath
- After some time, saturated NaCl solution added to precipitate the product
- Solid product scooped out with spatula, washed with water and then compressed into a small block of soap

Question 31 (b) (ii)

Criteria	Marks
Provides an explanation of the processes linking the structure of soap to its ability to form an emulsion in water with grease	4
- Outlines the structure of soap - Outlines some emulsifying properties of soap	3
- Outlines some features of the structure and/or emulsifying properties of soap OR - Identifies the product of saponification and that an emulsion is formed	2
- Identifies the product of saponification OR - Identifies a feature of the structure of soap OR - Makes a correct statement about the cleaning properties of soap	1

Sample answer:

The product of saponification is soap. This has a non-polar hydrophobic tail and a polar, hydrophilic negatively charged head. Soap is effective at cleaning non-polar grease from surfaces. The non-polar tail dissolves in the non-polar grease.



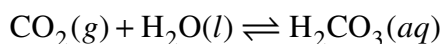
When enough soap is dissolved in the grease, the grease is lifted from the surface. The negatively charged heads of the soap prevent the grease drops from sticking together, effectively emulsifying the grease, allowing it to be carried away from the surface.

Question 31 (c) (i)

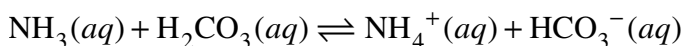
Criteria	Marks
- Clearly outlines the chemical processes that take place in the reaction vessel and the heating process - Includes chemical equations	3
- Outlines a chemical process that takes place in the production - Includes a chemical equation	2
Identifies a feature of a relevant chemical process	1

Sample answer:

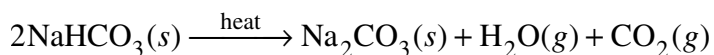
In the reaction vessel carbonic acid forms.



Ammonia converts this to the bicarbonate ion.



At low temperatures (0°C) Na^+ from the brine causes precipitation of $\text{NaHCO}_3(s)$. This is filtered and then heated to produce $\text{Na}_2\text{CO}_3(s)$.

**Question 31 (c) (ii)**

Criteria	Marks
Links the requirements for the location of a Solvay plant to relevant features of the diagram	3
Outlines some requirements for the location of a Solvay plant	2
- Identifies a requirement for the location of a Solvay plant OR - Identifies a suitable location for a Solvay plant	1

Sample answer:

In the diagram, substantial inputs include brine (often sourced from sea water) and carbon dioxide (obtained from limestone). Since ammonia is recycled, location of the plant is not dependent on ammonia production. CaCl_2 is the substantial by-product.

It is easier to transport bulk solid limestone long distances so it is more important to locate a Solvay plant near the ocean to access brine. Limestone deposits relatively close are desirable but not critical, so long as transport of limestone is available. Locating near the ocean also allows excess CaCl_2 to be disposed of easily into the sea without causing problems to marine ecology.

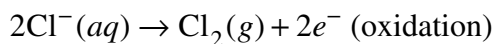
Question 31 (d)

Criteria	Marks
- Includes correct structure and chemistry of each method - Elaborates on technical and environmental issues associated with each method to show how the methods are similar and/or different - Provides a coherent and concise response with no extraneous information	7
- Includes correct structure and chemistry of each method - Elaborates on technical and environmental issues associated with each method	6
- Provides an outline and some chemistry of each method - Outlines technical and/or environmental issues associated with each method	4–5
Outlines some chemistry and/or technical issues(s) and/or environmental issues(s) for at least one of the methods.	2–3
- Includes some chemistry for one method OR - Names another method OR - Identifies a technical issue for one method OR - Identifies an environmental issue for one method	1

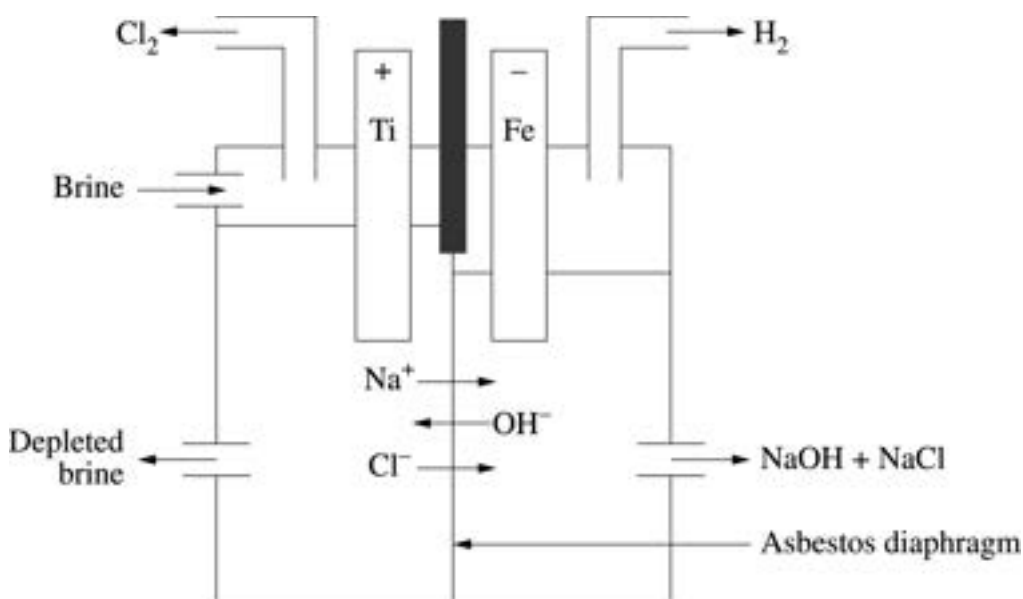
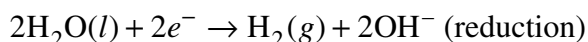
Sample answer:

Sodium hydroxide can be produced by the membrane cell method and a similar technique known as the diaphragm cell. The chemistry involved in each of these techniques is very similar, but there are significant technical and environmental differences between them.

In the diaphragm cell, brine is placed in a half-cell with a titanium anode where Cl^- ions are oxidised to Cl_2 .



The half-cell is separated from the cathode half-cell by a porous asbestos barrier. The iron mesh anode reduces molecules to OH^- and $\text{H}_2(g)$.



Problems associated with this method include the use of the asbestos diaphragm. There was the potential for workers to be exposed to carcinogenic asbestos during maintenance. In addition, migration of OH^- ions through the diaphragm produced ClO^- ions (a strong oxidising agent) into the depleted brine. This had to be removed prior to release into the environment. As well, Cl^- ions passing through the diaphragm would result in NaCl contamination in the final NaOH product.

As an alternative, the membrane cell uses identical chemistry to the diaphragm cell, but the anode and cathode compartments are separated by a semipermeable membrane produced from polytetrafluoroethylene (PTFE or Teflon). This allows positively charged sodium ions from the brine to pass through from the anode to the cathode cell to produce NaOH . However, negatively charged Cl^- and OH^- ions are repelled by the PTFE membrane. This prevents contamination of the depleted brine with hypochlorite and the contamination of NaOH with NaCl . In addition, the PTFE is harmless. Hence, the membrane cell is superior as it allows uncontaminated NaOH to be produced compared to the diaphragm method. It also prevents release of ClO^- into the depleted brine, and prevents risk of exposure of workers to asbestos.

Question 32 (a) (i)

Criteria	Marks
• Outlines relevant limitations	2
• States a relevant limitation	1

Sample answer:

Paint must be constantly reapplied to maintain the ship due to scratches and other damage and this might be very time consuming.

Question 32 (a) (ii)

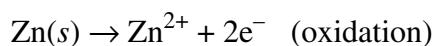
Criteria	Marks
• Provides an explanation of the chemical principles of sacrificial protection	3
• Includes relevant equations	
• Demonstrates some understanding of sacrificial protection	2
• Provides some relevant information	1

Sample answer:

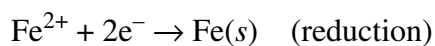
Metals that are more active than iron (Mg and Zn are commonly used) are attached to the iron in order to set up a galvanic cell. These metals are corroded away in preference to the iron and need to be regularly replaced.

For example:

The Zn is oxidised at the anode.



The electron flow into the iron prevents the formation of Fe^{2+} ions. The electrons produced by the oxidation of the zinc reduce any Fe^{2+} ions that form back to Fe atoms.



Question 32 (b) (i)

Criteria	Marks
- Provides characteristics and features of a relevant, valid and reliable first-hand investigation - Identifies equipment AND reagent used	3
- Outlines a relevant, valid and/or reliable first-hand investigation - Identifies equipment and/or reagent used	2
Identifies some steps of a relevant investigation	1

Sample answer:

1. Put a sample of iron (nail) and steel (stainless) of the same size in separate containers
2. Add a reagent such as salt water, of same concentration, to each sample so that each sample is half-submerged
3. Repeat steps 1–2 for 3 sets of samples
4. Label each container
5. Leave these samples to corrode and observe them over a few weeks

Question 32 (b) (ii)

Criteria	Marks
Relates the properties of TWO types of steel to their percentage composition	4
- Identifies the properties and percentage compositions of two types of steel - OR - Relates the properties of a type of steel to its percentage composition	3
- Identifies a property and the percentage composition of a type of steel - OR - Identifies the properties or percentage compositions of two types of steel - OR - Provides a relationship between property and percentage composition of steel	2
Identifies a property or the percentage composition of a type of steel	1

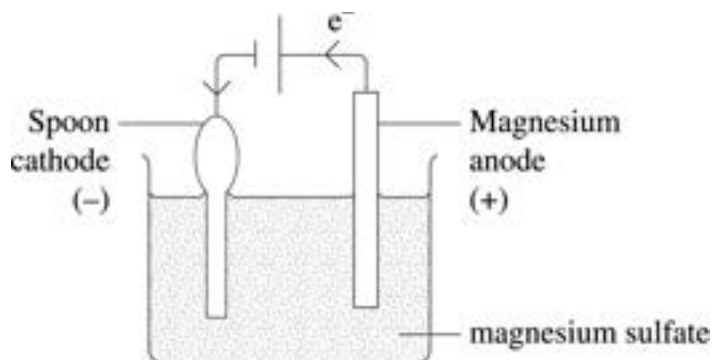
Sample answer:

Type of steel	Composition	Properties
Mild steel	Fe and < 0.2% carbon	Soft and malleable
Structural steel	Fe and > 0.3% to 0.6% carbon	Hard, high tensile strength

The lower the carbon content of the steel the more flexible and malleable it becomes.

Question 32 (c) (i)

Criteria	Marks
Provides a correctly labelled scientific diagram that includes the cathode, anode, direction of electron flow and polarity of the electrodes	3
Provides a scientific diagram with at least TWO of the following correctly labelled: cathode, anode, direction of electron flow, polarity of the electrodes	2
Provides a diagram with a cathode, anode or direction of electron flow	1

Sample answer:

Question 32 (c) (ii)

Criteria	Marks
• Clearly links Davy's work to the increased understanding of electron transfer reactions	3
• Outlines some features of Davy's work and/or electron transfer reactions	2
• Identifies a feature of Davy's work or electron transfer reactions	1

Sample answer:

Davy experimented with electrolytic cells, using much larger versions of Volta's pile, that produced much larger currents which were required to electrolyse more active metals. He discovered that active metals could be prepared by electrolysis of their molten salts and that aqueous electrolytes were not the only substances that could be decomposed by electrolysis. Davy was aware that charged particles were needed to be transported for electrolysis to occur. Davy also predicted large scale use of electrolysis to make alkalis and isolating elements.

Question 32 (d)

Criteria	Marks
- Explains how the rusting processes of the two wrecks differ by considering a range of relevant factors - Includes relevant equations - Provides a coherent and concise response with no extraneous information	7
- Provides a range of relevant factors that affect the rusting of the two wrecks - Explains how the rusting processes of the two wrecks differ by considering different factors - Includes at least one relevant equation	6
- Provides relevant factors that affect the rusting of the two wrecks - Outlines the rusting processes of the two wrecks in terms of at least one of the factors - Provides at least one relevant equation	4–5
- Provides factors that affect rusting - Outlines how at least one factor affects rusting and/or provides some relevant chemistry	2–3
- Identifies a feature of the rusting process OR - Provides a relevant equation OR - Identifies a relevant difference in the two environments	1

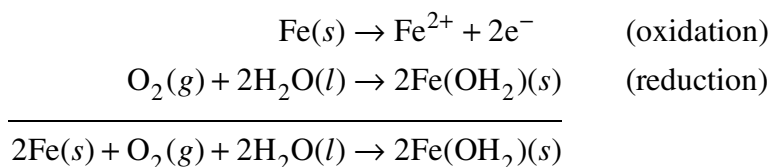
Sample answer:

The conditions affecting rusting are

- Presence of water
- Presence of dissolved oxygen
- The pH of the electrolyte
- The temperature
- A salty environment

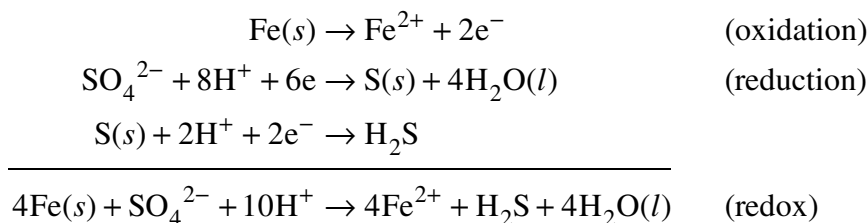
For sunken ships, the rate of decay and corrosion depends on the final depth of the wreck. Shallow wrecks corrode in the same way as objects on the surface, while wrecks in the deep ocean are corroded by bacterial action.

Shipwrecks in shallow water corrode as they are always wet, in contact with a good electrolyte with effective conductivity, and well aerated by wave motion allowing oxygen to dissolve.



With increasing depth of the ocean, pressure increases while temperature decreases. The solubility of gases increases with decreasing temperature and with increasing pressure. The higher the temperature, the harder it is for the gas to remain in solution. The higher the pressure, the more gas is compressed and will dissolve in water. In deep sections of the ocean, the concentration of dissolved oxygen is very low and as a result, corrosion should not occur or at least should be greatly retarded. Also the temperatures deep in the ocean are very low and corrosion reactions would proceed at a very slow rate.

Observations of deep ocean wrecks such as the Titanic show that corrosion is actually taking place. Bacteria causing corrosion on deep wrecks are sulfate-reducing species that produce acidic environments around the metals.



The acidic environments produced by sulfate-reducing bacteria accelerate corrosion in non-passivating metals such as iron.

Question 33 (a) (i)

Criteria	Marks
• Correctly identifies both components with correct respiration type	2
• Correctly identifies a component OR • Correctly identifies a respiration type	1

Sample answer:

Component	Respiration type
Mitochondria	Oxidative decarboxylation
Cytoplasm	Glycolysis

Question 33 (a) (ii)

Criteria	Marks
• Clearly relates the biological significance of ATP to its structure	3
• Provides the function of ATP and its structure OR • Provides some link between the function of ATP and its structure	2
• Provides the structure of ATP OR • Identifies the function of ATP	1

Sample answer:

ATP has the following structure



The bonds between the phosphates provide a large amount of energy when broken to convert ATP to ADP. $\text{ATP} \rightarrow \text{ADP} + \text{P}$.

The energy released in this reaction is used to drive metabolic reactions.

Question 33 (b) (i)

Criteria	Marks
- Provides characteristics and features of a relevant, valid and reliable first-hand investigation - Identifies equipment AND reagent used	3
- Outlines a relevant valid and/or reliable first-hand investigation - Identifies equipment and/or reagent used	2
Identifies some steps of a relevant investigation	1

Sample answer:

- Set up a range of water baths from 0°C to 100°C
- Place 1.0 mL of 3% H₂O₂ in a number of identical test tubes
- Place two test tubes in each of the water baths and allow them to reach the temperature of the water bath
- Add the same amount of crushed potato (pea size) to one test tube in each water bath. Leave other as a control
- Measure the height of the bubbles produced as a result of the action of the catalase on H₂O₂
- Repeat the procedure three times for reliability.

Question 33 (b) (ii)

Criteria	Marks
- Recognises the role of models in visualising key enzyme function at cellular level - Relates the shape of the enzyme to its specificity - Provides advantages and/or disadvantages of using models to understand how enzymes function in living systems	4
- Identifies way(s) that models can be used to understand how enzymes function - Identifies advantage(s) and/or disadvantage(s) of using models to understand how enzymes function	3
- Identifies way(s) that models can be used to understand how enzymes function OR - Identifies advantage(s) and/or disadvantage(s) of using models to understand how enzymes function	2
- Identifies a way that models can be used to understand how enzymes function OR - Identifies an advantage/disadvantage of using models to understand how enzymes function OR - Shows a basic understanding of how enzymes function	1

Sample answer:

Enzymes are catalysts, a special class of proteins that speed up cellular chemical reactions at their active sites. Use of models helps visualise the nature of these enzymes and their specificity.

Models are generally used to explain the enzyme-substrate complex that provides a low energy pathway necessary to catalyse a specific reaction.

The active site of the enzyme has a specific shape (similar to a lock) and only the correctly shaped substrate (similar to a key) will fit that site enabling bond-making and bond-breaking.

A model such as 'lock and key' is too static and does not explain the conditions which are necessary to bring the reactants together, such as collision of molecules, effect of temperature, pH and so on. It also has a fixed shape (a template) that is not true of enzymes. They have a 3D structure. However, the use of models also allows a complex biological process to be more easily understood at the molecular level.

Question 33 (c) (i)

Criteria	Marks
• Relates TWO bonding types from the diagram to the shape of the protein	3
• Provides TWO bonding types OR • Relates ONE bonding type to the shape of the protein	2
• Provides some relevant information	1

Sample answer:

As shown in the diagram there are HN and $\text{C}-\text{O}-\text{H}$ groups on adjacent amino acids (but not neighbouring ones) that have formed hydrogen bonds between them, responsible for holding the folded pleat in place thus giving secondary structure to the protein.

The $-\text{CH}_2\text{NH}_3^+$ and $\text{O}^- - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 -$ have electrostatic forces of attraction between them thus enabling the adjacent chain to form a fold. This bond gives strength to the structure and renders shape.

Question 33 (c) (ii)

Criteria	Marks
• Outlines how changes in conditions can disrupt the intermolecular forces leading to denaturing of this protein making it biologically inactive	3
• Outlines how a change in conditions can disrupt the intermolecular forces leading to denaturing this protein	2
• Identifies ONE relevant feature of denaturation	1

Sample answer:

Denaturing a protein such as the one shown in the diagram involves disrupting its secondary and tertiary structure thus making the protein biologically inactive.

Factors that can cause denaturing are changes in pH and temperature. The protein then coagulates into solid lumps.

Changes in pH (eg adding acid) will cause the $\text{O}^- - \overset{\text{O}}{\parallel} \text{C} -$ group to become $\text{HO} - \overset{\text{O}}{\parallel} \text{C} -$. This will destroy the electrostatic attraction between the NH_3^+ and O^- .

An increase in temperature will increase the kinetic energy, disrupting these relatively weak bonds.

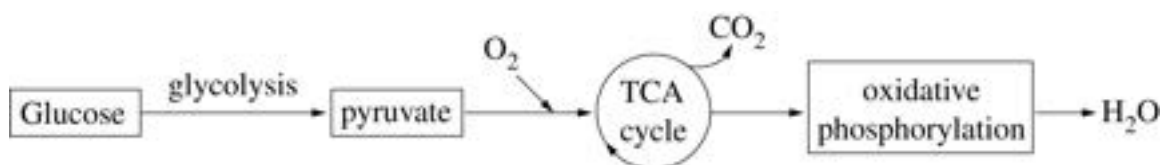
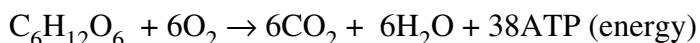
Question 33 (d)

Criteria	Marks
- Shows how the metabolic pathways involved in sprinting and those involved in gentle walking are similar and/or different - Considers a range of areas such as muscle fibres, rate of ATP production, fuels used, oxygen availability and the wastes produced - Provides a coherent and concise response with no extraneous information	7
- Provides the metabolic pathways involved in sprinting and gentle walking - Compares these pathways by considering areas such as muscle fibres, rate of ATP production, fuels used, oxygen availability and the wastes produced	6
- Provides metabolic pathways involved in sprinting and gentle walking - Elaborates on areas such as muscle fibres, ATP production, fuels used, oxygen availability and waste produced in relation to these pathways	4–5
- Provides metabolic pathway(s) involved in sprinting and/or gentle walking AND/OR - Recognises the role of glucose breakdown (glycolysis)	2–3
Identifies a feature related to sprinting or general walking	1

Sample answer:

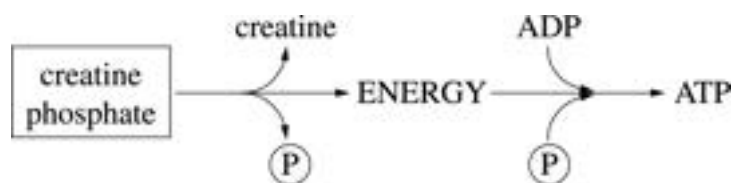
During gentle exercise, under aerobic conditions, Type 1 muscles (slow twitch muscle fibres) are involved. They have a rich blood supply that enables them to obtain oxygen in plenty. Glucose is metabolised in glycolysis to produce pyruvate that proceeds to the TCA cycle due to plentiful supply of oxygen. This is followed by oxidative phosphorylation. The total energy output of aerobic respiration is 38 ATP.

Other than glucose in the blood, amino acids and fatty acids and glycerol could also be used as fuels for aerobic respiration. Carbon dioxide and water are the waste products of glucose metabolism that are eliminated easily.

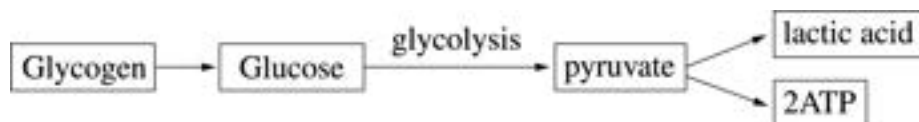


A sprinting athlete uses predominantly Type 2 muscles (fast twitch muscle fibres). These muscle fibres have a poor blood supply and as a result are deficient in oxygen. These muscles respire under anaerobic conditions when very limited or no oxygen is available. Sprinting involves intense exercise and needs large bursts of energy in a very short period of time. There are two systems that rapidly provide the energy required.

- Phosphocreatine system (PCr) The energy release is small but at a very rapid rate. Most of the energy in a creatine phosphate molecule is in the phosphate bond. Breaking the phosphate bond releases energy that is used by the athlete in sprinting. ATP is generated in this process.

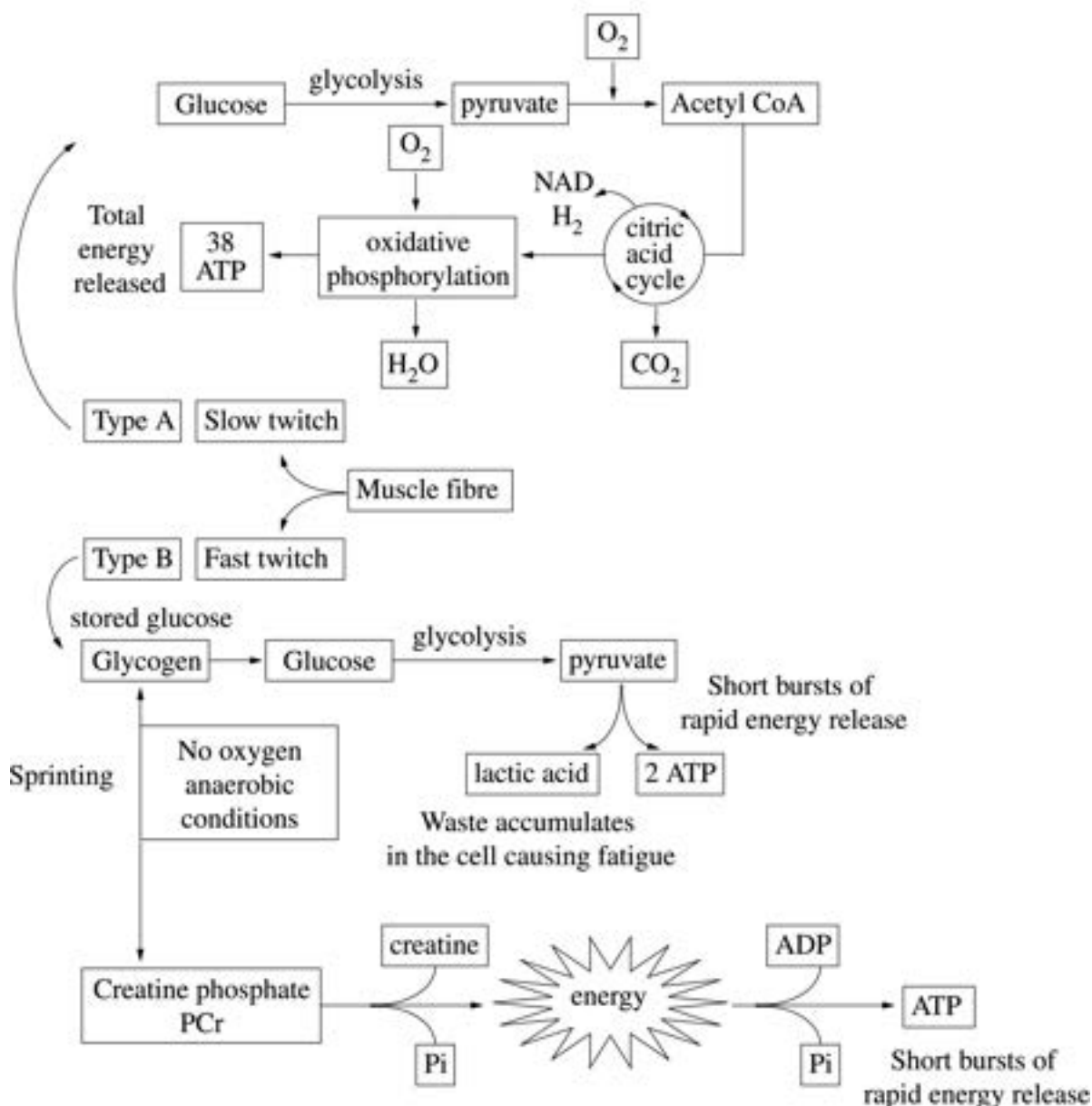


- ii. Anaerobic glycolysis The fuel for anaerobic glycolysis comes from stored glycogen in muscle fibres. Due to low glucose concentrations in blood, glycogen is instantly converted into glucose that is then converted into pyruvate. Because of lack of oxygen, the pyruvate does not proceed to the TCA cycle, instead producing short bursts of energy plus lactic acid which accumulates in the cells as waste causing muscle fatigue.



Answers could include:

Slow walking Aerobic conditions



Question 34 (a) (i)

Criteria	Marks
- Provides the chemical composition of a cosmetic used in an ancient culture - Identifies its potential health risk	2
- Identifies a toxic component of a cosmetic OR - Identifies a health risk associated with the use of cosmetics	1

Sample answer:

Cinnabar (HgS) was used in lipstick and contains mercury. This neurotoxin is easily absorbed through the skin.

Question 34 (a) (ii)

Criteria	Marks
- Relates the colour of Cr^{2+} to its electron configuration and hence the splitting of the d orbital - Relates the lack of colour in Zn^{2+} to the lack of transition within the d orbital	3
- Relates the colour of Cr^{2+} to its electron configuration and hence the splitting of the d orbital OR - Relates the lack of colour in Zn^{2+} to the lack of transition within the d orbital	2
Provides some relevant information	1

Sample answer:

Cr^{2+} has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$. It has unpaired d subshell electrons and, when surrounded by water ligands as $\text{Cr}^{2+}(\text{aq})$, the orbitals of its 3d subshell are split in energy. Absorption of some wavelength/s of light causes an electron/s to be excited to the higher energy orbitals and hence the solution appears coloured. In comparison, $\text{Zn}^{2+}(\text{aq})$ is not coloured since it does not have any unpaired d subshell electrons. It has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$.

Question 34 (b) (i)

Criteria	Marks
Provides a detailed procedure for a safe and valid investigation, identifying the specific metal ions observed	3
- Includes the majority of the steps and the relevant metal ions that can be observed OR - Provides a detailed procedure for a relevant, safe and valid investigation	2
- Identifies a metal ion that can be identified using a flame test OR - Includes an appropriate chemical OR - Provides a relevant step OR - Identifies a hazard OR - Includes a safety measure	1

Sample answer:

- Safety glasses should be worn due to the corrosive nature of concentrated HCl
- A nichrome wire was dipped into some concentrated HCl to clean it
- The wire was then dipped into a small amount of one of the following salts: sodium nitrate, potassium nitrate, calcium nitrate, barium nitrate, copper(II) nitrate and strontium nitrate
- The nichrome wire was then placed into the hottest part of the Bunsen flame and the colour of the flame noted
- The wire was cleaned by dipping into the concentrated HCl and the process repeated until the flame colour of all salts had been observed.

Question 34 (b) (ii)

Criteria	Marks
- Provides an explanation showing clear relationship between cause and effect - Names one metal that cannot be identified using flame tests	4
- Provides an explanation showing some link between cause and effect, without naming one metal that cannot be identified using flame tests OR - Names one metal that cannot be identified using flame tests and provides some reasons why only certain metal ions can be identified using flame colours	3
- Names one metal that cannot be identified using flame tests and identifies a reason why only certain metal ions can be identified using flame colours OR - Identifies reasons why only certain metal ions can be identified using flame colours	2
Provides some relevant information	1

Sample answer:

Flame colours are produced by the movement of electrons in the metal ions found in the compound. When the metal ion is heated, the electrons gain energy and jump to an orbital at a higher energy level. Because the electrons are now at a higher and more energetically unstable level, they fall back down to their original energy level, emitting electromagnetic radiation, often seen as light of a certain wavelength. Not all metal ions can be detected using flame tests, as electrons in certain metals, in returning to their original energy level, emit wavelengths which are not detectable by the human eye because they may be emitted in the ultraviolet and infrared regions of the electromagnetic spectrum. Aluminium is one such metal ion that releases wavelengths of energy outside the range visible to humans.

Answers could include:

Relationship between

- heating a metal ion and the movement of electrons to higher energy levels
- the return of electrons to their ground state and the emission of energy
- wavelengths emitted and visibility

Question 34 (c) (i)

Criteria	Marks
- Identifies the reduced IE for Al and S - Relates the lower first IE for Al and S to Hund's rule and the existence of orbitals	3
- Recognises some trend in IE across Period 3 of the Periodic Table and/or identifies the reduced IE for Al and S OR - Links the lower first IE for Al and S to Hund's rule and/or the existence of orbitals	2
- Recognises some trend in IE across Period 3 of the Periodic Table OR - Identifies the reduced IE for Al and S OR - Shows a basic understanding of Hund's rule or the existence of orbitals	1

Sample answer:

In general, first IE increases across the third period. However, Al has a lower 1st IE compared to Mg. In Mg, the two valence electrons are both in the s-subshell. In Al, the third valence electron is in a p-subshell, which is at a higher energy than the s-subshells, hence, less energy is required to remove the outermost electron in Al than in Mg. Phosphorous has 5 valence electrons, with two in the s-subshell and one in each of 3 p-subshells, according to Hund's rule. Sulfur has 6 valence electrons, requiring one of the p-subshells to have two electrons in it. Electrostatic repulsion between these two electrons results in less energy required to remove a valence electron, reducing the 1st IE of S compared to P.

Question 34 (c) (ii)

Criteria	Marks
• Uses the graph to identify the trends in electronegativity both down a group and across a period • Relates changes in electronegativity to the arrangement of electrons	3
• Identifies the trend(s) in electronegativity down a group and/or across a period AND/OR • Links changes in electronegativity to the arrangement of electrons in atoms	2
• Shows a basic understanding of electronegativity OR • Identifies the trend in electronegativity down a group or across a period OR • Recognises a link between electronegativity and the arrangement of electrons	1

Sample answer:

Electronegativity is a measure of the tendency of an atom to attract electrons to itself. As can be seen from the graph, electronegativity increases across a period from left to right and decreases down a group. Usually, the inert gases are not considered, because they do not usually form bonds and hence F is the most electronegative element. Electronegativity increases across a period due to an increase in nuclear charge, attracting bonding electrons more strongly. Although there is also an additional electron added with each element, this does not fully shield the effect of the increased number of protons. Electronegativity decreases down a group, because, with the addition of an extra electron shell, the size of the atom increases, the bonding pair of electrons is increasingly distant from the nucleus and the attractive force of the shielded nucleus is diminished.

Question 34 (d)

Criteria	Marks
- Provides the relevant features of the Bohr model of the atom - Provides strengths and weaknesses of the model in understanding the atomic structure and makes an informed judgement implicitly/explicitly - Makes reference to emission spectra - Provides a coherent and concise response with no extraneous information	7
- Provides the relevant features of the Bohr model of the atom - Provides strengths and/or weaknesses of the model in understanding the atomic structure and makes an implicit/explicit judgement	6
- Provides some features of the Bohr model of the atom - Provides strength(s) and/or weakness(es) of the Bohr model in understanding the atomic structure	4–5
Provides some features of the Bohr model of the atom and/or atomic structure and/or emission spectra	2–3
Identifies a feature of the Bohr model of the atom or the atomic structure	1

Sample answer:

Bohr proposed the ‘shell’ model of the atom, in which he envisaged electrons moving around the nucleus in orbits of fixed size and energy. Part of the reason why Bohr developed this model was to answer some of the unsatisfactory aspects of the Rutherford “planetary” model, which could not explain how charged particles could move in a circular path, emitting electromagnetic radiation and yet not spiral into the nucleus.

Bohr’s model accounts for the fact that atoms emit or absorb energy when electrons change energy levels within the atom. This explained the observation that excited atoms generate line emission spectra, when the electrons jump to a higher energy level and then return to their original level; the energy difference being released as a specific frequency of electromagnetic radiation. For the hydrogen spectrum, this model worked well. However, attempts to extend the Bohr model to atoms containing more than one electron, including helium with a more complex structure, gave results which did not agree with the line spectrum wavelengths observed. Although the model was useful in explaining some of the lines for the helium spectrum, it was too simplistic to explain them all. The model could also not explain that upon detailed analysis of the hydrogen spectrum, a single emission line was actually composed of two or more closely spaced lines. Moreover, it could not explain the Zeeman effect – changes in spectral lines due to external magnetic fields.

Certainly, there are several fundamental weaknesses associated with the Bohr model. However, it did contribute to our knowledge of atomic structure and has since had a number of enhancements made to it in the development of the Bohr-Sommerfeld model.

Question 35 (a) (i)

Criteria	Marks
• Shows how the compositions of glycogen and cellulose are similar and/or different	2
• Identifies a feature of glycogen's or cellulose's composition	1

Sample answer:

Both polymers are made of glucose monomers. Glycogen is made of α -glucose, cellulose is made of β -glucose.

Question 35 (a) (ii)

Criteria	Marks
• Correctly relates glycogen's composition to its structure • Correctly relates cellulose's composition to its structure	3
• Links glycogen's composition to its structure OR • Links cellulose's composition to its structure OR • Identifies features of glycogen's and cellulose's structures	2
• Identifies a feature of glucogen's or cellulose's structure	1

Sample answer:

Glycogen – due to α linkages forms branching side chains so it cannot align closely together and so forms a network.

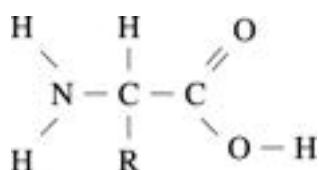
Cellulose – due to β linkages forms linear, unbranched molecules, which align closely together forming hydrogen bonds between chains making it strong and rigid.

Question 35 (b) (i)

Criteria	Marks
- Identifies the structure of amino acids - Describes the relationship between amino acids and proteins	3
Identifies features of the structure of amino acids and/or the relationship between amino acids and proteins	2
- Identifies a feature of the structure of amino acids OR - Identifies a feature of the relationship between amino acids and proteins	1

Sample answer:

Amino acids consist of a central carbon to which is attached a carboxyl group, an amine group, a hydrogen atom and a variable side chain.



Amino acid monomers join together by peptide bonds to form long polypeptide chains. The final 3D structure of the functional protein depends on the nature of the amino acid side chains.

Question 35 (b) (ii)

Criteria	Marks
- Correctly describes a procedure that: - shows the presence of protein including results - addresses validity - outlines how a risk is minimised	4
- Describes a substantially correct procedure that: - identifies the presence of protein including results - addresses validity and risk	3
Outlines a procedure that identifies the presence of protein, addresses validity, identifies a risk or includes some expected results	2
Provides some relevant information	1

Sample answer:

- Take two test tubes and add 1 mL egg white to one and 1 mL of water to the other.
- Add 10 drops of Biuret reagent A ($2.0 \text{ mol L}^{-1} \text{ NaOH}$) to each sample.
- Then add 10 drops of Biuret reagent B ($0.10 \text{ mol L}^{-1} \text{ CuSO}_4$) to each sample.
- Agitate and warm gently.
- The test tube containing water (the control) remains pale blue, the test tube containing protein becomes purple/violet in colour.

Safety glasses must be worn as Biuret reagent A (NaOH) is corrosive.

Question 35 (c) (i)

Criteria	Marks
Identifies the relationships of the children A, B and C to George and Linda with appropriate justification	3
- Identifies the relationships of the children A, B and C to George and Linda OR - Identifies some relationships with some justification	2
Provides some relevant information	1

Sample answer:

The parents of child A are George and Linda because A's bands match both George and Linda's bands.

George must be a parent of child B because some of B's bands match George's bands but none of them match Linda's.

Linda must be a parent of Child C because some of C's bands match Linda's bands but none of them match George's.

Question 35 (c) (ii)

Criteria	Marks
Provides a description of the benefits of maintaining DNA data banks	3
- Outlines some benefits of maintaining DNA data banks OR - Describes a benefit of maintaining DNA data banks	2
Identifies a feature or a benefit of DNA data banks	1

Sample answer:

DNA data banks collect and store DNA profiles (fingerprints) from many individuals mostly from volunteers or convicted criminals. They contain very sensitive and accurate information and allow identification of individuals very precisely (1 in 6 billion). They can allow fast and accurate identification of perpetrators. They can quickly and definitively prove somebody innocent of a crime such as rape if no DNA match.

Question 35 (d)

Criteria	Marks
- Names ONE chromatography technique - Provides the relevant features of the technique - Provides strengths and weaknesses of its use in the analysis of forensic evidence and makes an informed judgement implicitly/explicitly - Provides a coherent and concise response with no extraneous information	7
- Names ONE chromatography technique - Provides the relevant features of the technique - Provides strengths and/or weaknesses of its use in the analysis of forensic evidence and makes an implicit/explicit judgement	6
- Names ONE chromatography technique - Provides some features of the technique - Provides strengths and/or weaknesses of its use in the analysis of forensic evidence	4–5
Provides some features of chromatography techniques in the analysis of forensic evidence	2–3
Provides some relevant information	1

Sample answer:

GLC – Gas Liquid Chromatography

This type of chromatography separates substances on the basis of their different distributions between two phases. These are either a stationary phase (usually a porous solid) and a mobile phase, which is usually an inert gas, that passes over the stationary phase. The stationary phase is packed into a column which can be 1–10 m long. The column is housed inside an oven and the temperature of the oven is set to be above the boiling points of the components of the mixture. The sample is injected into the mobile phase which carries the vapours into and through the column. As the vapours make contact with the stationary phase, some components condense more readily than others. As the mobile phase moves over the condensed components, some evaporate and are carried out of the column to a detector. Each individual component has its own unique retention time.

GLC is a very sensitive technique that can measure quantities in ppb range. The analysis can be done very quickly so that results are quickly generated. It can also be used to analyse blood and urine samples to detect illegal drugs and alcohol. It is quicker and cheaper than many other techniques and produces detailed and precise measurements from very small amounts of material. If GLC is coupled to a mass spectrometer the analysis can identify components accurately and is highly reliable for court evidence. Thus GLC is an extremely accurate and reliable method of analysing forensic evidence.

2015 HSC Chemistry Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.4.4.2.1; 9.4.4.2.2; 9.4.4.2.3	H6
2	1	9.3.4.3.2; 9.3.4.2.8	H6
3	1	9.4.3.3.1	H6
4	1	9.2.4.2.3	H6
5	1	9.3.2.2.2	H6
6	1	9.4.4.2.10	H9
7	1	9.2.4.2.4	H6
8	1	9.2.3.2.3	H9
9	1	9.3.5.2.4	H9
10	1	9.2.1.3.1; 9.4.1.2.3	H9
11	1	9.2.2.2.3	H9
12	1	9.2.5.2.1	H6
13	1	9.3.3.2.5; 9.3.3.2.6; 9.3.3.2.7	H6; H12.4 (b);
14	1	9.3.4.3.3	H6; H11.3 (a); H14.1 (a)
15	1	9.4.5.2.1; 9.4.5.2.2	H6; H14.1 (a)
16	1	9.3.2.2.3; 9.3.2.2.4	H7; H8; H10
17	1	9.2.3.3.4; 9.2.2.9	H6; H9; H10; H12.4 (b)
18	1	9.4.5.3.3	H12.1 (a); H12.4 (e)
19	1	9.4.5.3.1; 9.4.5.3.2	H10; H12.4 (b)
20	1	9.2.3.2.7	H9; H12.4 (b); H14.1 (a)

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.3.1.3.1, H11.2 (c)	H11
21 (b)	2	9.3.1.3.1, H11.2 (c)	H11
22 (a)	4	9.4.4.2.11	H13.1 (f); H13.1 (g)
22 (b)	3	9.4.4.3.1; 9.4.4.2.1; 9.4.4.2.11	H3; H4
23	4	9.2.4.3.3	H3; H4
24 (a)	2	9.3.4.2.4	H6; H8; H13.1 (d); H14.1 (g)
24 (b)	3	9.3.4.2.9	H6; H8; H13.1 (d); H14.1 (g)
25 (a)	3	9.2.1.2.3; 9.2.1.2.4; 9.2.1.2.5; 9.2.1.2.6	H9
25 (b)	4	9.2.1.2.8	H3; H9
26 (a)	1	9.3.3.2.2	H9; H13.1 (d)
26 (b)	2	9.3.4.3.4	H9; H11.3 (c); H12.2 (a)
26 (c)	4	9.3.4.3.3	H10; H12.4 (b)

Question	Marks	Content	Syllabus outcomes
27	5	9.2.5.2.5; 9.2.5.2.6; 9.2.5.3.2	H1; H3; H4
28 (a)	1	9.3.4.2.1	H8
28 (b)	2	9.3.4.2.2	H8; H13.1 (d)
29 (a)	4	9.4.3.3.4	H8; H10; H12.4 (e)
29 (b)	3	9.4.3.3.3	H8; H10; H12.4 (b)
30	6	9.4.2.2.2; 9.4.2.2.3; 9.4.2.2.4; 9.4.2.2.5; 9.4.2.2.6; 9.4.2.2.7; 9.4.2.2.8; 9.4.2.2.9; 9.4.2.2.10	H7; H8; H10

Section II

Question	Marks	Content	Syllabus outcomes
Question 31		Industrial Chemistry	
31 (a) (i)	2	9.5.2.2.1	H6; H10
31 (a) (ii)	3	9.5.2.2.2; 9.5.2.3.3	H6; H8; H10; H12.4 (b)
31 (b) (i)	3	9.5.5.3.1; 9.5.5.2.1	H6; H9; H11.3 a); H12.1 (a); H12.2 (b)
31 (b) (ii)	4	9.5.5.2.3; 9.5.5.2.4	H6; H9;
31 (c) (i)	3	9.5.6.2.3	H6; H7
31 (c) (ii)	3	9.5.6.2.1; 9.5.6.3.3	H6; H7
31 (d)	7	9.5.4.2.2; 9.5.4.2.3	H6; H7; H8
Question 32		Shipwrecks, Corrosion and Conservation	
32 (a) (i)	2	9.6.4.3.3, 9.6.4.3.4	H10
32 (a) (ii)	3	9.6.4.2.4, 9.6.4.3.4	H10
32 (b) (i)	3	9.6.2.3.1	H6; H11.3 (a); H12.1 (a); H12.2 (b)
32 (b) (ii)	4	9.6.2.2.3	H6; H8
32 (c) (i)	3	9.6.1.3.1	H6; H8; H13.1 (e)
32 (c) (ii)	3	9.6.1.2.4	H1; H2
32 (d)	7	9.6.5.2.1; 9.6.5.2.3; 9.6.5.2.3; 9.6.6.2.1; 9.6.6.2.2; 9.6.5.3.1; 9.6.5.3.2	H6; H7; H8
Question 33		The Biochemistry of movement	
33 (a) (i)	2	9.7.5.2.1	H1
33 (a) (ii)	3	9.7.1.2.2	H2
33 (b) (i)	3	9.7.4.3.2	H4
33 (b) (ii)	4	9.7.4.3.3	H2
33 (c) (i)	3	9.7.4.2.5	H1, H6
33 (c) (ii)	3	9.7.4.2.6	H1, H6
33 (d)	7	9.7.8.3, 9.7.10.2.1, 9.7.10.3.1	H7, H8, H9
Question 34		The Chemistry of Art	
34 (a) (i)	2	9.8.1.3.2; 9.8.1.3.3	H6
34 (a) (ii)	3	9.8.4.2.3; 9.8.4.2.4; 9.8.5.3.2	H6

Question	Marks	Content	Syllabus outcomes
34 (b) (i)	3	9.8.2.2.1; 9.8.2.3.1	H6; H11.3 (a); H12.1 (a); H12.2 (b)
34 (b) (ii)	4	9.8.2.2.2; 9.8.2.2.3	H6; H11.2 (d)
34 (c) (i)	3	9.8.3.3.1	H6; H14.1 (a); H14.1 (f)
34 (c) (ii)	3	9.8.3.2.7	H6; H14.1 (a); H14.1 (f)
34 (d)	7	9.8.2.3.4; 9.8.2.2.5	H1; H2; H6; H7
Question 35		Forensic Chemistry	
35 (a) (i)	2	9.9.2.2.2	H9
35 (a) (ii)	3	9.9.2.2.4	H9; H14.3 (d)
35 (b) (i)	3	9.9.3.2.2; 9.9.3.2.3	H9; H14.3 (d)
35 (b) (ii)	4	9.9.3.3.2	H12.1 (b); H12.1 (d); H12.2 (b)
35 (c) (i)	3	9.9.4.2.2; 9.9.4.2.3	H4; H14.2 (c)
35 (c) (ii)	3	9.9.4.3.1	H4
35 (d)	7	9.9.3.2.5; 9.9.5.2.2; 9.9.5.3.1	H3; H11.3 (c)